GREENHOUSE GAS ANALYZER AND GPS DATA SHEET

FILE NAMES

Lake: Falls Lake

Sample Date: 10/23/18

GGA file_2018-10-23_f0001.txt

gps file name

DEPLOYMENT DATA

Sample site	Chamber start time	Bubbling observed (Y/N)	GPS logger started	Chamber volume (L)	CH_4 and CO_2 profile checked	Water Temp (degC)
53	8:51:50	N	y	45	y	14.8
42	9:01:47	N	✓	45	✓	15
50	9:11:31	N	✓	45	✓	15
44	9:18:43	N	✓	40	✓	15.1
52	9:29:49	y	✓	40	✓	16.1
51	9:30:48	N	✓	40	✓	16.1
56	9:42:00	N	✓	40	✓	16.1
47	9:49:26	N	✓	40	✓	16.3
43	9:57:57	y	✓	40	✓	16.0
43	10:01:25	N	✓	40	✓	15.9
54	10:09:15	y	✓	40	✓	16.1
54	10:11:48	N	✓	40	✓	16.2
55	10:19:34	N	✓	40	✓	16.3
41	10:26:44	N	✓	40	✓	16.5
49	10:34:43	N	✓	40	✓	16.6
57	10:46:07	N	✓	40	✓	17.1
45	10:53:01	N	✓	40	✓	17.5
10	11:00:05	N	✓	40	✓	17.8
4	11:12:00	N	✓	40	✓	17.7
8	11:33:50	N	✓	40	✓	17.7
3	11:44:38	N	✓	40	✓	18.2
14	12:01:41	N	✓	40	✓	18.5
15	12:16:24	N	✓	40	✓	18.8
11	12:24:48	N	✓	40	✓	18.7
7	12:41:30	y	✓	40	✓	19.7
1	12:55:38	N	✓	40	✓	20.2
21	13:06:12	y	✓	40	✓	20
21	13:12:37	y	✓	40	✓	19.9
21	13:14:55	N	✓	40	✓	19.9
13	13:25:50	N	✓	40	✓	20.3
12	13:44:37	N	✓	40	✓	20.9

5	13:57:00	N	✓	40	✓	21.0
2	14:09:13	y	✓	40	✓	21.2
2	14:12:11	N	✓	40	✓	21.2
16	14:26:08	N	✓	40	✓	21.6
24	14:40:03	N	✓	40	✓	21.7

Signature

10/23/18
Date

CO_2 and CH_4 should go up steadily once chamber is on water, CO_2 can go down in high algae scenarios

SONDE DATA SHEET

Lake: FAlls Sample Date: 10/22/18

Site	Site Depth (m)	Sample Depth (m)	Temp (C)	DO (mg/l)	DO (%sat)	sp.Cond ($\mu s\ cm^{-1}$)	pH	Chl a (ug/L)	Turbidity (NTU)	Notes
53	1.3	0.1	15.1	11.16	110.7	0.173	7.06	32.2	15.7	
42	1.2	0.1	15.0	10.49	103.9	0.145	6.96	34.6	17.7	
42	1.2	0.7	15.1	10.10	100.3	0.145	7.01	33.0	21.0	
50	1.1	0.1	15.2	10.56	104.7	0.092	7.14	27.8	24.8	
44	1.2	0.1	15.4	11.18	111.5	0.105	7.14	32.1	19.7	
44	1.2	0.7	14.9	9.9 ~~10.53~~	98.2 ~~104.5~~	0.080	7.29	28.4	26.2	
52	1.4	0.1	16.6	10.57	107.4	0.089	7.22	26.0	18.6	
56	2.5	0.1	16.9	9.62	99.1	0.083	7.31	27.0	15.0	
47	2.2	0.1	16.4	9.96	101.4	0.084	7.39	27.0	16.3	
43	1.9	0.1	16.1	10.20	103.5	0.084	7.42	22.2	18.6	
43	1.9	1.4	15.9	9.7	98.0	0.084	7.42	23.9	42.0	
54	2.0	0.1	16.7	9.89	100.9	0.084	7.41	21.0	17.0	
55	2.4	0.1	16.5	10.30	105.4	0.081	7.38	24.0	15.7	
41	1.1	0.1	16.7	10.45	107.2	0.081	7.41	20.8	14.5	
41	1.1	0.6	16.1	10.30	104.4	0.081	7.44	31.2	16.9	
49	2.8	0.1	16.7	10.40	107.3	0.081	7.45	28.4	13.4	
57	2.3	0.1	17.9	10.30	108.5	0.082	7.42	16.7	10.3	
45	2.6	0.1	18.3	9.48	100.7	0.082	7.47	22.3	10.9	
10	1.4	0.1	18.2	9.57	101.5	0.082	7.47	21.6	11.5	
4	1.2	0.1	18.0	10.24	108.2	0.082	7.42	18.4	10.5	
4	1.2	0.7	17.7	9.98	104.7	0.080	7.47	22.5	13.8	
8	2.9	0.1	18.8	10.17	109.0	0.082	7.55	16.2	8.4	
3	1.5	0.1	18.4	10.25	109.0	0.081	7.50	19.4	8.9	
3	1.5	1.0	17.3	9.60	99.4	0.081	7.51	22.0	11.0	
14	1.6	0.1	18.5	9.84	104.4	0.076	7.35	11.2	7.3	
14	1.6	1.0	18.0	8.49	89.4	0.076	7.47	14.6	14.0	
15	4.5	0.1	19.5	10.19	110.2	0.078	7.46	14.9	6.7	
15	4.5	4.0	18.7	8.09	86.5	0.080	7.49	14.2	8.6	

Signature

10/23/18
Date

SONDE DATA SHEET

Lake: FAllS

Sample Date: 10/22/18

Site	Site Depth (m)	Sample Depth (m)	Temp (C)	DO (mg/l)	DO (%sat)	sp.Cond (µs cm^{-1})	pH	Chl a (ug/L)	Turbidity (NTU)	Notes
11	3.6	0.1	19.3	9.71	105.6	0.080	7.49	16.3	6.6	
7	7.8	0.1	20.2	7.86	97.1	0.082	7.37	14.5	7.3	
7	7.8	1.0	19.9	7.77	85.5	0.082	7.41	17.0	7.8	
7	7.8	2.0	19.7	6.76	73.6	0.082	7.46	12.4	8.2	
7	7.8	3.0	19.6	6.49	70.8	0.082	7.45	12.4	8.1	
7	7.8	4.0	19.6	6.43	70.3	0.082	7.44	10.8	8.7	
7	7.8	5.0	19.6	6.48	70.7	0.082	7.42	13.0	7.8	
7	7.8	6.0	19.3	6.50	70.5	0.082	7.41	11.0	9.6	
7	7.8	7.0	18.8	6.48	69.5	0.082	7.38	10.5	14.0	
7	7.8	7.3	18.7	6.43	68.9	0.082	7.34	16.8	32.4	
1	5.1	0.1	20.8	7.68	85.6	0.084	7.20	11.6	5.3	
1	5.1	4.6	20.2	5.6	61.9	0.082	7.22	9.8	9.6	
21	2.7	0.1	20.1	8.45	93.2	0.087	7.31	17.2 [illegible]	10.0	
13	4.0	0.1	21.0	8.03	90.2	0.082	7.29	13.5	5.9	
12	4.5	0.1	21.6	7.98	90.4	0.085	7.18	11.0	4.2	
5	4.6	0.1	21.2	7.18	80.9	0.083	7.29	12.0	6.5	
5	4.6	4.1	20.6	5.35	59.3	0.083	7.26	8.6	8.0	
2	3.7	0.1	21.1	8.85	99.6	0.092	7.30	8.7	5.6	
2	3.7	3.2	19.3	8.60	95.8	0.093	7.11	8.5	6.5	
16	6.5	0.1	22.4	7.65	87.9	0.085	7.21	9.0	4.7	
24	2.3	0.1	21.7	6.14	69.1	0.085	7.23	8.2	4.8	

Signature

10/23/18

Date

DISSOLVED GAS AND AIR DATA SHEET

Lake:	Falls				Date:	10/23/2018				
*Site	Dissolved gas Exetainer codes Prefix: 1022298.FL.		Water Vol/ ~~Helium~~ Air Vol	Water Temp (degC)	Atmospheric Air Exetainer Codes Prefix: 10222018.FL.					BP (mm Hg)
42	10222018.FL.42.DG.01	42.DG.02	115mL/25mL	14.8						
44	44.DG.01	44.DG.02		15.1						
43	43.DG.01	43.DG.02		15.9						
41	41.DG.01	41.DG.02		16.5	41.AA.01	41.AA.02	41.AA.03	41.AA.04	41.AA.05	
04	04.DG.01	04.DG.02		17.7						
03	03.DG.01	03.DG.02		18.2						
14	14.DG.01	14.DG.02		18.5						
15	15.DG.01	15.DG.02		18.8						
07	07.DG.01	07.DG.02		19.7						
01	01.DG.01	01.DG.02		20.2						
05	05.DG.01	05.DG.02		21.0						
02	02.DG.01	02.DG.02		21.2	02.AA.01	02.AA.02	02.AA.03	02.AA.04	02.AA.05	

*Site: all dissolved gas samples collected at 0.1m depth

Boat motor should be off for 2 minutes before sample collection

Goal is to inject 20mL of gas into exetainer

10/23/2018

Signature

Date

GAS TRAP DATA SHEET

Lake: FALLS

Site	Lat	Long	Deployment time/date Local 10/22/18	Retrieval time/date 10/23/18	Volume recovered (mL)	Exetainer codes
53	36. 069000	78. 794756	0915	0840	0	
42	36. 065237	78. 795192	0922	0853	10.5	
50	36. 068143	78. 786059	0931	0900	0.5	
44	36. 072857	78. 779209	0939	0910	25,18	10222018.FL.44.BT.01/ FL.44.BT.02
52	36.079635	78. 760329	0953	0921	0.5	
56	36.072773	78. 764282	1001	0935	25,25,25,4	10222018.FL.44.BT.01/BT.02/BT.03
47	36. 068287	78. 768611	1008	0940	17	10222018.FL.47.BT.01
43	36. 061452	78. 773379	1017	0949	25, 16.5	10222018.FL.43.BT.01
54	36. 057763	78. 765154	1027	1000	2.5	
55	36.053050	78. 761547	1035	1010	2	
41	36. 055760	78.758572	1041	1017	25	
49	36. 051225	78.756767	1049	1026	8	
57	36. 048438	78.752466	1100	1037	5	
45	36. 043960	78. 758244	1107	1040?	4.5	
10	36. 039748	78. 756087	1113	1050	–	
4	36. 012138	78. 761056	1125	1103	0.5	
8	36. 019040	78. 748576	1137	1123	23.5	10222018.FL.08.BT.01
3	36. 032483	78. 736996	1146	1144?	0.5	
14	36. 059522	78. 717112	1206	1155	0.5	
15	36. 022717	78. 714644	1219	1207	1.5	
11	36. 019297	78. 717119	1232	1215	0.5	
7	36. 005468	78. 667906	1248	1232	25, 7	10222018.FL.07.BT.01
1	36. 002120	78. 634477	1313	1247	1	
21	36.008950	78. 628192	1320	1255	9	
13	36. 003617	78. 646282	1330	1316	0.5	
12	35. 965570	78. 633922	1345	1338	25, 1	10222018.FL.12.BT.01
5	35. 970052	78. 621271	1352	1347	0.5	
2	35. 976380	78. 588947	1405	1400	13	
16	35. 956753	78. 590414	1420	1419	25, [illegible]	10222018.FL.16.BT.01
24	35. 941995	78. 588159	1430	1430	–	

Signatur

10/23/18

Date